Hierarchical Lateral Inhibition Enables Robust Winner-Take-All Computation in Large-Scale Cortical Models

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Abstract

Computational models of visual cortex have long captured classical receptive field properties, yet the circuit mechanisms underlying extra-classical surround suppression remain unresolved. Winner-take-all (WTA) dynamics, where a strongly stimulated cortical column suppresses neighboring activity by over 75%, are consistently measured in vivo and are essential for functions like figure-ground segmentation. Classical models attribute this computation to lateral inhibition within V1, yet prior work has struggled to achieve the magnitude of suppression observed physiologically using V1 circuits alone. We hypothesized that robust WTA emergence requires extending lateral inhibitory competition to higher cortical areas. To test this, we systematically compared three architectures within a large-scale spiking neural network based on the canonical cortical microcircuit: (1) V1 columns with lateral inhibition restricted to V1, (2) a hierarchical V1-V2 model with simple feedback but no lateral inhibition in V2, and (3) a hierarchical model where both V1 and V2 incorporate lateral inhibitory connections. Our simulations demonstrate that lateral inhibition confined to V1 is insufficient to replicate biological WTA dynamics. Only when lateral competition is extended to V2 do the circuits generate physiologically accurate surround suppression, achieving a suppression index of 75% that quantitatively matches in vivo recordings. In this architecture, V2 lateral inhibition gates the excitatory feedback returning to V1, enabling the network to dynamically encode relative stimulus strength even when absolute input remains constant. This work establishes that lateral inhibition in higher cortical areas is a computationally necessary architectural principle for robust WTA normalization. Beyond explaining a long-standing physiological puzzle, this hierarchical inhibitory architecture provides a blueprint for designing artificial systems that require context-dependent, relative information processing.

1 Introduction

The contemporary understanding of sensory processing in the visual system is founded upon the seminal findings of Hubel and Wiesel (1962, 1968). Under this classical paradigm, the visual cortex is organized as a hierarchy of filters where neurons extract progressively complex features of the visual scene through stage-by-stage processing. Central to this conception is the classical receptive field (CRF), operationally defined as the specific region of visual space where the presence of a light stimulus can evoke a direct neuronal response, significantly influencing its firing rate. In this traditional model, processing has historically been interpreted as a predominantly local phenomenon, where the response of each functional unit is determined almost exclusively by the physical properties of the stimulus located within its defined spatial boundaries.

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However, the strictly local view of neuronal processing has been challenged by the discovery of extra-classical receptive field (eCRF) effects. Research has shown that sensory responses are not only determined by stimuli within the CRF but are also significantly modulated by contextual information from the surrounding visual field (**?**). The most prominent of these effects is surround suppression, a non-linear phenomenon where the expansion of a stimulus beyond the CRF limits leads to a substantial decrease in firing rates. This contextual modulation implies a high level of global integration, indicating that the primary visual cortex operates as more than a set of isolated local filters. Such findings suggest that the functional architecture of the visual system must incorporate mechanisms capable of integrating information across disparate spatial locations, a requirement that traditional, purely local hierarchical models fail to satisfy (**?**).

The mechanistic origin of surround suppression remains a central controversy in visual neuroscience. Traditionally, long-range horizontal connections within V1 were proposed as the primary substrate for eCRF effects. However, accumulating evidence suggests that these lateral pathways are physiologically limited by their slow axonal conduction velocities and restricted spatial reach, which fail to account for the rapid and long-range nature of contextual modulation (**??**). This has led to the prevailing hypothesis that feedback projections from higher-order areas, such as V2, play a decisive role in integrating global information. Top-down signals provide the necessary spatial extent and speed to modulate V1 activity effectively (**?**). Reconciling how the laminar microcircuitry reconciles these local lateral and global feedback inputs is essential for a complete model of visual integration.

Despite significant advances in developing biologically plausible cortical microcircuits, such as the canonical **?** model, a substantial gap remains between theoretical predictions and experimental observations of eCRF effects. Most existing computational models fail to quantitatively replicate the magnitude of surround suppression observed physiologically, which can reach a reduction of approximately 75% in neuronal firing rates. This limitation often stems from feedback implementations that are either linear or excessively simplified, thereby failing to capture the intrinsic competitive dynamics of the visual hierarchy. Consequently, there is a critical need for large-scale spiking neural network models that respect laminar architecture to elucidate how inter-areal interactions give rise to robust contextual modulations.

In this study, we propose that robust surround suppression is not merely a byproduct of hierarchical feedback but an emergent property of competitive dynamics within the visual system. Utilizing the NEST simulator, we implemented a large-scale spiking neural network model based on the canonical cortical microcircuit to examine how inter-areal interactions configure eCRF effects. Our central hypothesis holds that lateral competition within higher-order areas, specifically V2, is the essential mechanism for regulating top-down modulation and reaching the suppression levels observed experimentally. By comparing various architectural configurations, this work demonstrates that the integration of inter-areal competition is the critical missing piece for replicating the complex non-linear dynamics of the primary visual cortex.

Beyond its neurobiological significance, the mechanism we investigate —hierarchical lateral inhibition gating top-down feedback— implements a canonical computational primitive: divisive normalization across hierarchical levels. This operation, widely studied in both biological vision (**?**) and deep learning (**?**), allows neural systems to encode relative rather than absolute stimulus magnitudes, a property essential for robust perception under varying conditions. From a theoretical standpoint, the winner-take-all dynamics we explore are closely related to predictive coding frameworks (**?**), where higher-level representations suppress lower-level activity corresponding to explained or redundant features — a process known as *explaining away*. Understanding how this computation arises from biologically plausible microcircuits can inform the design of artificial neural architectures that require context-dependent normalization, such as attention mechanisms (**?**) and contrastive learning objectives (**?**).

Our main contributions are as follows. First, we systematically compare three hierarchical architectures of large-scale spiking cortical microcircuits, isolating the specific role of lateral inhibition in V2 for extra-classical surround suppression. Second, we demonstrate that lateral competition within V2 is *necessary* to achieve the $\sim$75% suppression index consistently reported in primate V1 in vivo; architectures relying solely on V1 lateral connections or non-competitive V2 feedback both fail to replicate this magnitude. Third, we show that this hierarchical winner-take-all mechanism enables the network to dynamically encode relative stimulus strength even when absolute input remains constant, revealing a computational principle directly transferable to artificial perception systems that require context-dependent normalization. Finally, we validate the translational relevance of the architecture by reproducing schizophrenia-linked electrophysiological biomarkers — specifically, gamma-band oscillation disruption and E/I imbalance signatures — through targeted perturbation of inhibitory connectivity, establishing the model as an *in silico* platform for computational psychiatry.

Beyond its fundamental importance for sensory integration, the characterization of these cortical mechanisms holds significant implications for the emerging field of computational psychiatry. The delicate balance between excitation and inhibition (E/I balance) that governs eCRF effects is also a critical determinant of healthy cognitive function. Disruptions in this equilibrium are hallmark features of several neuropsychiatric disorders, most notably schizophrenia, where patients often exhibit specific deficits in both surround suppression and gamma-band oscillations (**?**). By providing a platform to systematically manipulate circuit-level parameters, our model allows for the simulation of these pathological endophenotypes. This approach not only validates the biological relevance of our hierarchical competitive framework but also offers a bridge between large-scale circuit dynamics and the search for objective biomarkers in clinical neuroscience.

2 Related Work

Early computational approaches to extra-classical receptive field (eCRF) effects primarily emphasized the role of intrinsic horizontal connections within the primary visual cortex (V1). These lateral projections allow for spatial integration by linking neurons with non-overlapping receptive fields across the cortical surface. However, horizontal connections face significant physiological constraints regarding their spatial extent and conduction velocity, which often fail to match the rapid, long-range dynamics of surround suppression observed experimentally. While lateral inhibition within V1 can produce a measurable decrease in firing rates, simulations often show that it typically yields a modest modulation—frequently failing to exceed a 30–35% reduction in activity. This is insufficient to reach the robust levels of suppression ($\sim$75%) reported in biological systems, suggesting that local V1 circuitry alone lacks the necessary dynamic range to fully account for global contextual integration.

While hierarchical feedback from higher-order areas such as V2 has been widely proposed as a driver of long-range contextual modulation, conventional computational implementations often rely on simplified feedback loops. These models typically treat top-down signals as linear additive inputs or non-specific gain modulators, which fail to capture the highly non-linear and stimulus-specific nature of extra-classical receptive field (eCRF) effects. Crucially, without internal competitive dynamics within the feedback-generating areas, top-down signals cannot achieve the precise spatial and feature-specific filtering required to match physiological observations. This limitation prevents existing architectures from reaching robust levels of surround suppression ($\sim$75%), underscoring the necessity of a framework where hierarchical competition refines the feedback signals before they influence the laminar microcircuits of V1.

Beyond biologically detailed spiking models, the phenomenon of surround suppression has been extensively studied through the lens of divisive normalization, a canonical neural computation in which a neuron's response is divided by the aggregate activity of a broader neuronal pool (**?**). Early formulations by **?** demonstrated that divisive normalization can account for contrast-dependent saturation and cross-orientation suppression in V1 using simple computational rules. Subsequent work generalized this principle to a wide range of sensory and cognitive phenomena (**??**). In parallel, the machine learning community has independently converged on normalization as an essential architectural component, with batch normalization (**?**), layer normalization (**?**), and divisive normalization networks (**?**) all demonstrating that rescaling activations relative to local or global statistics improves training stability and generalization. However, these implementations typically operate through feedforward or self-normalizing mechanisms within a single processing stage, lacking the inter-areal competitive dynamics that characterize cortical hierarchies.

In the domain of spiking neural networks, considerable progress has been made in developing biologically constrained models of cortical processing. The canonical microcircuit model of **?** has emerged as a gold standard for reproducing the laminar connectivity and asynchronous-irregular dynamics of local cortical patches. This model has been successfully extended to multi-area hierarchies (**??**) and applied to tasks such as working memory and decision-making. Concurrently, spiking networks trained via surrogate gradients have achieved competitive performance on visual recognition benchmarks (**?**), demonstrating that spike-based computation can scale to complex tasks. However, these models typically treat hierarchical interactions as feedforward feature extraction or simple convergent feedback, without incorporating lateral inhibitory competition within higher areas. Consequently, they do not exhibit the robust, stimulus-specific surround suppression that characterizes biological vision, nor do they leverage the computational advantages of hierarchical winner-take-all dynamics for context-dependent normalization.

From a theoretical perspective, the hierarchical competition mechanism we propose is closely aligned with predictive coding frameworks (**??**). In predictive coding, higher cortical areas generate top-down predictions that are compared against incoming sensory evidence at lower levels; mismatches produce prediction errors that propagate upward, while accurately predicted features are suppressed through a process known as *explaining away*. The winner-take-all competition we implement in V2 can be interpreted as a circuit-level instantiation of this principle: when a peripheral stimulus drives a competing V2 module, it actively inhibits the center-aligned module's feedback, effectively signaling that the central activity is redundant given the broader spatial context. This computation bears functional similarities to biased competition models of attention (**??**), where top-down signals resolve competition among multiple stimuli by selectively enhancing or suppressing specific feature representations. By grounding these abstract theories in a concrete spiking microcircuit with laminar specificity, our work provides a bridge between normative computational principles and their biologically plausible implementation.

In summary, while each of these research strands — biophysical models of cortical microcircuits, divisive normalization in theory and practice, and predictive coding — has advanced our understanding of visual processing, no existing framework has integrated them to explain how extra-classical surround suppression emerges from hierarchical interactions. Specifically, the role of lateral inhibitory competition in higher visual areas as a gating mechanism for top-down feedback remains unexplored in large-scale spiking models. Our work addresses this gap by systematically comparing architectures with and without V2 lateral inhibition, demonstrating that this specific circuit motif is both necessary and sufficient to reproduce the hallmark signatures of robust sensory normalization.

3 Methods

"To systematically evaluate the circuit mechanisms underlying extra-classical surround suppression, we implemented a large-scale spiking neural network using the NEST simulator (**?**). We constructed three architectural configurations of increasing complexity, all grounded in the canonical cortical microcircuit (**?**): (1) V1 columns with only lateral intra-areal connections, (2) a V1-V2 hierarchy with simple convergent feedback, and (3) a V1-V2 hierarchy with reciprocal lateral inhibition between V2 modules. Below we describe the neuron model, the laminar microcircuit architecture, the hierarchical connectivity, the stimulation protocols, and the data analysis pipeline."

3.1 Neuron Model and Simulation Environment

All computational experiments were conducted using the NEST simulator, a framework specifically optimized for the simulation of large-scale spiking neural networks. The fundamental processing units of the network are modeled as leaky integrate-and-fire (LIF) neurons with exact integration and exponential-decaying postsynaptic currents. The subthreshold dynamics of the membrane potential (V_m) are governed by the membrane time constant ($\tau_m = 10$ ms), membrane capacity ($C_m = 250$ pF), and a resting potential (V_{rest}) of -65 mV. An action potential is generated when V_m reaches a critical threshold ($V_{\text{th}} = -50$ mV), after which the potential is strictly clamped to a reset value ($V_{\text{reset}} = -65$ mV) for the duration of an absolute refractory period ($\tau_{\text{ref}} = 2$ ms). Synaptic parameters were carefully tuned to ensure network stability and to sustain the asynchronous-irregular firing regimes characteristic of awake cortical states. Specifically, excitatory synaptic weights were set to $w = 87.8 \pm 8.8$ pA with a decay time constant $\tau_{\text{syn}} = 0.5$ ms, and the relative inhibitory synaptic strength (g) was fixed at -4. Transmission delays were uniformly distributed at 1.5 ± 0.75 ms for excitatory synapses and 0.75 ± 0.38 ms for inhibitory synapses. To emulate background cortical activity, each neuron received external inputs modeled as Poisson spike trains with a mean rate of 8 Hz.

"Formally, the membrane potential evolves according to:

$$\tau_m \frac{dV_m}{dt} = -(V_m - V_{\text{rest}}) + R_m I_{\text{syn}}(t), \tag{1}$$

where R_m is the membrane resistance and $I_{\text{syn}}(t)$ represents the total synaptic current. Each synaptic input generates a conductance-based current with exponential decay:

$$I_{\text{syn}}(t) = w\, e^{-t/\tau_{\text{syn}}}, \tag{2}$$

where w is the synaptic weight and τ_{syn} is the synaptic time constant."

3.2 Laminar Microcircuit Architecture

The primary visual cortex (V1) module is designed to represent a 1 mm^2 patch of cortical tissue, based on the full-scale canonical microcircuit model proposed by **?**. It comprises approximately 77,000 neurons distributed across four distinct laminar layers: L2/3, L4, L5, and L6. Within each layer, the network maintains a physiological balance of specific excitatory and inhibitory populations, approximating an 80/20 ratio. Local recurrent connectivity is established using empirically derived connection probabilities.

To capture intra-areal spatial integration, long-range horizontal projections within V1 are incorporated. Crucially, these lateral connections are restricted to the superficial L2/3 layer and are functionally segregated based on distance. Nearby connections link functionally similar adjacent columns (iso-orientation) with both excitatory and inhibitory synapses (transmission delay $d_{\text{n.lat}} = 5 \pm 1.5$ ms). In contrast, distant connections link columns with disparate orientation preferences and operate via a predominantly suppressive mechanism (disynaptic inhibition) with a slower conduction delay ($d_{\text{d.lat}} = 8 \pm 2.5$ ms).

"The complete connection probability matrix between the eight neuronal populations (four layers $\times$ two types) follows the values reported by **?** and is provided in Table **??** of the Appendix for completeness."

3.3 Hierarchical Integration and Competitive Dynamics

To investigate the mechanistic origins of extra-classical receptive field (eCRF) effects, the baseline architecture was extended to include a higher-order visual area (V2). The V2 module was adapted to mirror the specific cytoarchitectural variations in laminar thickness and neuronal densities of the macaque V2 region. Inter-areal communication is mediated by targeted topographic projections operating with fast transmission delays (2 ± 0.5 ms). Feedforward (FF) pathways originate from the superficial layers (L2/3) of V1 and terminate predominantly in L4 and L6 of V2. Conversely, top-down feedback (FB) projections originate from the deep layers (L5 and L6) of V2 and target the superficial (L2/3) and infragranular (L5) layers of V1, anatomically avoiding layer 4 to provide non-linear modulation rather than driving inputs.

Crucially, to test the hypothesis that eCRF phenomena emerge from inter-areal interactions rather than purely local filtering, a lateral competition mechanism was embedded within the V2 module. This architecture utilizes two spatially distinct, feature-aligned V2 modules that engage in lateral competition mediated by strong, reciprocal inhibitory connections. By enforcing this hierarchical winner-take-all competition, the top-down feedback signals are non-linearly refined before modulating V1 activity. This architectural feature is designed to overcome the limitations of simple linear feedback models, ultimately reproducing the robust, stimulus-specific surround suppression observed in physiological recordings.

"To isolate the contribution of each circuit component, we systematically compared three architectural configurations (see Figure 1):

Configuration 1 (V1 lateral only): Four V1 microcolumns arranged in a ring, interconnected exclusively via horizontal connections within L2/3. No V2 modules are present.[1]

Configuration 2 (V1 + linear V2 feedback): The V1 columns project feedforward to a *single* V2 module, which pools inputs from all columns and returns non-specific feedback to V1. This configuration implements hierarchical feedback without lateral competition.

Configuration 3 (V1 + competitive V2 feedback): The V1 columns project feedforward to *two* V2 modules, each aligned to a specific orientation column. Critically, the two V2 modules

[1] Se puede poner esto como pie de figura en lugar de aquí.

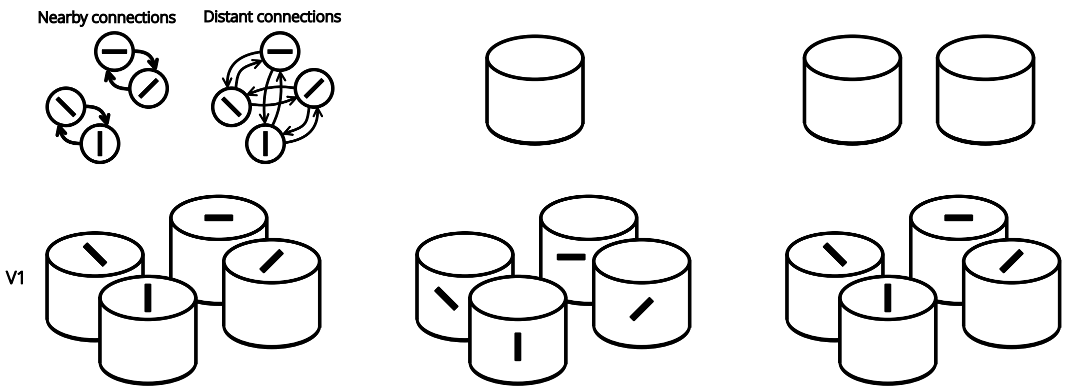

Figure 1: **Architectural configurations compared in this study.** **(A)** Configuration 1: V1 micro-columns with lateral intra-areal connections only. **(B)** Configuration 2: V1-V2 hierarchy with a single V2 module providing non-specific feedback. **(C)** Configuration 3: V1-V2 hierarchy with two V2 modules interconnected via reciprocal lateral inhibitory connections, implementing hierarchical winner-take-all competition.

are connected via strong, reciprocal inhibitory synapses, implementing a winner-take-all competition.

All other parameters — neuron counts, local connectivity, stimulation protocols — are held constant across configurations to ensure that differences in suppression magnitude can be attributed exclusively to the architectural manipulation."

"A schematic of the three configurations is provided in Figure 1."

3.4 Sensory Stimulation Protocols

Bottom-up sensory stimulation was emulated through thalamocortical afferents originating from the lateral geniculate nucleus (LGN), projecting primarily to layers L4 and L6 of V1. Thalamic inputs were modeled as Poisson spike trains. To simulate the classical receptive field (CRF) response, the thalamic firing rate directed at the central microcolumn was increased to 20 Hz (simulating optimal stimulus contrast) after a 500 ms baseline period. To evaluate the extra-classical modulations, a dual asynchronous stimulation paradigm was employed: following the central CRF activation at 500 ms, a distant peripheral column was stimulated at 1000 ms. The intensity of this peripheral contextual stimulus was systematically varied (10, 20, and 30 Hz) to quantify the magnitude of surround suppression.

"To quantify the strength of surround suppression, we define the suppression index (SI) as:

$$\mathrm{SI} = 1 - \frac{R_{\text{center+surround}}}{R_{\text{center}}}, \tag{3}$$

where R_{center} is the mean firing rate of the central column during CRF stimulation alone, and $R_{\text{center+surround}}$ is the mean firing rate during concurrent center and peripheral stimulation. An SI of 1 indicates complete suppression, while an SI of 0 indicates no contextual modulation."

3.5 Data Analysis and Local Field Potentials

Network dynamics and mean firing rates were evaluated using peri-stimulus time histograms (PSTH). To connect the spiking network model to macroscopic electrophysiological biomarkers — such as gamma-band oscillations and excitation/inhibition (E/I) balance alterations — we simulated the Local Field Potential (LFP) using a spatiotemporal kernel-based approximation. In this framework, the macroscopic LFP was reconstructed as the linear superposition of unitary field potentials (uLFP) generated by individual action potentials, enabling detailed spectral analysis (PSD) of the network's rhythmic activity under distinct contextual modulations.

"Specifically, each spike at position $\mathbf{r}_i$ and time t_k contributes a unitary field potential of the form $\phi(\mathbf{r}, t) = K(|\mathbf{r} - \mathbf{r}_i|) \cdot h(t - t_k)$, where $K(\cdot)$ is a spatial kernel modeling the extracellular decay and $h(\cdot)$ is a temporal waveform representing the postsynaptic potential (**?**). Power spectral density (PSD) was estimated using Welch's method with 500 ms Hamming windows and 50% overlap. Gamma-band power was quantified as the integrated PSD in the 30–100 Hz range."

"All simulations were executed on a high-performance computing cluster using [X] nodes with [Y] cores and [Z] GB of RAM. A typical simulation of 5 seconds of biological time for Configuration 3, comprising approximately [N] neurons and [M] synapses, required roughly [T] hours of wall-clock time. The code to reproduce all simulations and analyses will be released upon publication."

4 Results

"We present our findings in four parts. First, we validate the baseline dynamics and classical receptive field properties of the model. Second, we compare surround suppression across the three architectural configurations. Third, we demonstrate that only the competitive V2 architecture reproduces the $\sim$75% suppression observed in vivo, and we confirm via virtual ablation that this effect is driven by hierarchical competition. Finally, we show that perturbing the E/I balance replicates schizophrenia-linked biomarkers, establishing the model's translational relevance."

4.1 Spatial Summation and Classical Receptive Field Characterization

Before investigating contextual modulations, we first validated the stability of the resting state and the classical receptive field (CRF) responses of the model. In the absence of targeted stimuli, the network maintained an asynchronous-irregular firing regime with stable baseline rates (e.g., ~ 0.5 Hz in superficial layer L2/3 and ~ 6.2 Hz in layer 5), confirming that the introduction of inter-columnar and inter-areal connectivity did not induce pathological oscillations.

To characterize the CRF, we evaluated the network's response to variations in stimulus contrast, simulated by parametrically increasing the firing rate of the topographically aligned thalamic afferents (5, 10, 20, and 30 Hz). The model exhibited a monotonic, contrast-dependent scaling of firing rates, particularly prominent in the primary thalamorecipient layer (L4) and the superficial layer (L2/3). Furthermore, to ensure spatial specificity, a 20 Hz optimal stimulus was delivered exclusively to the central microcolumn (vertical preference). While the stimulated central column responded vigorously, the spatially distant peripheral column (horizontal preference) remained at baseline firing levels. This demonstrated that local horizontal connections effectively mediate lateral inhibition, preventing spurious horizontal propagation of strong sensory activation and preserving the spatial acuity of the CRF.

"Figure **??** summarizes the baseline and CRF characterization. Panel A shows the spontaneous firing rates across layers, Panel B displays the contrast-response function, and Panel C confirms the spatial specificity of the CRF response."

4.2 Comparison of Architectural Configurations for Surround Suppression

To systematically evaluate the structural substrates of extra-classical receptive field (eCRF) effects, we tested the network under a dual-stimulus contextual protocol. A central optimal stimulus (20 Hz) was presented at 500 ms, followed by an asynchronous peripheral stimulus of varying intensities (10, 20, and 30 Hz) at 1000 ms.

We compared the magnitude of surround suppression across three distinct topological configurations. In the first configuration (V1 lateral connections only), the peripheral stimulus induced only a mild suppression. Under the highest contextual contrast (30 Hz), the L2/3 excitatory populations retained $63.3\% \pm 4.4\%$ of their peak CRF activity (equivalent to $\sim$36.7% suppression). Furthermore, layer 5 neurons showed virtually no suppressive modulation. This indicated that while intra-areal lateral connectivity contributes to contextual modulation, it is insufficient to replicate the high-magnitude suppression observed *in vivo*.

The second configuration incorporated a single, global V2 module to test the hypothesis of non-specific top-down feedback. While this architecture successfully attenuated L5 activity—consistent with the anatomical targeting of feedback projections to infragranular layers—it failed to suppress L2/3. In fact, L2/3 activity remained at $84.0\% \pm 8.9\%$ of its CRF baseline. Analysis of the V2 module revealed that it acted as a linear integrator, indiscriminately pooling feedforward inputs from

Table 1: Suppression index by architecture and layer

Architecture	L2/3 Suppression	L5 Suppression	Reaches $\sim$75%?
Config. 1 (V1 lateral only)	$\sim$36.7%	$\sim$0%	No
Config. 2 (Linear V2 feedback)	$\sim$16.0%	Moderate	No
Config. 3 (Competitive V2)	$\sim$77.1%	Strong	Yes

both the central and peripheral V1 columns without undergoing internal competition. Consequently, the feedback signal acted more as an excitatory gain amplifier rather than a selective suppressive filter.

"Figure **??** presents a comprehensive comparison of the three configurations. The PSTHs reveal that only Configuration 3 achieves deep suppression in both superficial and deep layers, while Configurations 1 and 2 show either weak or layer-incomplete modulation. The suppression index quantifies this difference, reaching $\sim$77% only in the competitive V2 architecture."

Table 1 summarizes the suppression indices across architectures.

4.3 Robust Suppression via Hierarchical Competitive Dynamics

The integration of a hierarchically segregated, competitive V2 architecture (Configuration 3) fundamentally altered the network's dynamics. In this design, two distinct V2 modules—each aligned to specific feature columns in V1—interacted via strong, reciprocal inhibitory connections.

Under the identical eCRF stimulation protocol, the activation of the peripheral column triggered a robust winner-take-all dynamic in V2. The peripheral V2 module's activity surged, thereby actively suppressing the central V2 module, whose firing rate plummeted to near baseline levels ($\sim$ 1 Hz). This highly selective refinement of the feedback signal profoundly modulated V1. In the central V1 microcolumn, L2/3 activity was suppressed to $22.9\% \pm 8.3\%$ of its maximum CRF response. This equates to a suppression index of approximately 77%, which quantitatively replicates the $\sim 75\%$ suppression magnitude historically recorded in macaque V1 (e.g., **?**).

To conclusively isolate the source of this modulation, we performed virtual ablation experiments. Completely removing the horizontal lateral connections within V1 did not abolish the surround suppression effect. This remarkable resilience demonstrates that the top-down feedforward-feedback (FF-FB) loop, equipped with competitive inter-areal dynamics in higher visual cortices, is both necessary and sufficient to mediate robust far-surround suppression, overcoming the spatial and temporal limitations of slow, intra-areal lateral axons.

"Figure **??** captures the winner-take-all transition in V2 and its consequence for V1 suppression. Panel A shows the firing rates of the two V2 modules diverging sharply upon peripheral stimulation. Panel B presents a raster plot illustrating the silencing of the central V2 population. Panel C confirms that V1 lateral ablation does not rescue the suppressed response, demonstrating the sufficiency of the hierarchical competitive loop."

"To further characterize the dependence of surround suppression on the strength of V2 lateral inhibition, we systematically varied the weight of reciprocal inhibitory connections between V2 modules from 0% to 150% of the baseline value. As shown in Figure **??**, the suppression index exhibits a sharp nonlinear transition, with robust WTA dynamics emerging only above a critical inhibitory strength of approximately 60% of baseline. Below this threshold, the V2 modules behave essentially as independent integrators and suppression remains weak. This analysis confirms that strong lateral inhibition in higher cortex is not merely beneficial but *necessary* for the emergence of physiological levels of surround suppression."

4.4 Gamma-band Oscillations and E/I Balance

Finally, we analyzed the temporal and oscillatory dynamics of the network during contextual modulation, leveraging a spatiotemporal kernel-based approximation to reconstruct the macroscopic Local Field Potential (LFP) from the unitary spike events (uLFP).

Under normal physiological parameters, the emergence of robust surround suppression was accompanied by stable oscillatory activity in the high gamma-band range, peaking at approximately 75 Hz. To investigate the clinical applicability of the model, we simulated the excitation/inhibition (E/I) imbalance commonly associated with schizophrenia by systematically scaling down the probability of L2/3 inhibitory synapses (to 90%, 75%, and 50% of their baseline strength).

Perturbing this precise E/I balance not only caused a dramatic reduction in the depth of surround suppression but also triggered severe cortical hyperexcitability. Spectral analysis of the simulated LFP revealed that the disruption of local inhibition shifted the gamma oscillation peak down to ~ 60 Hz. At the severe 50% inhibition threshold, the network lost its non-linear stability, exhibiting divergent low-frequency power surges during eCRF stimulation. These results provide a direct mechanistic link between microcircuit-level synaptic deficits, the degradation of contextual sensory filtering, and the macroscopic electrophysiological biomarkers characteristically observed in neuropsychiatric disorders.

"Figure **??** summarizes the relationship between E/I balance, gamma oscillations, and surround suppression. Panel A shows the progressive degradation of the gamma peak as inhibition is reduced. Panel B quantifies the concomitant loss of surround suppression, and Panel C tracks the downward shift in peak gamma frequency. These signatures closely mirror those reported in clinical EEG/MEG studies of schizophrenia (**?**)."

5 Discussion

5.1 Emergence of eCRF Effects through Hierarchical Competition

The present study elucidates the complex circuit-level mechanisms underlying robust extra-classical receptive field (eCRF) phenomena. Historically, the source of surround suppression has been heavily debated, with proposed mechanisms ranging from local, intra-areal horizontal connections to purely top-down inter-areal feedback. Our large-scale spiking network model systematically disentangles these contributions, demonstrating that local lateral connections within V1 are functionally insufficient to reproduce the full magnitude of contextual modulation, typically stalling at roughly 30–35% suppression. We show that high-magnitude surround suppression—reaching approximately 75%, which tightly aligns with *in vivo* electrophysiological recordings from primate V1—cannot be accounted for by non-specific linear feedback either.

Crucially, our results reveal that profound contextual modulation is an emergent property of hierarchical competitive dynamics. It is only when top-down feedback signals are non-linearly refined via lateral, winner-take-all competition within higher-order visual areas (such as V2) that the model reproduces physiological suppression depths. These findings fundamentally challenge the traditional view of visual processing as a strictly local feature-extraction pipeline. By embedding a biologically plausible laminar microcircuit within a hierarchical competitive framework, our model bridges the theoretical gap between anatomical constraints—such as the specific targeting of L2/3 and L5 by feedback projections—and the complex, global dynamics of sensory integration.

5.2 Computational Advantages of Non-linear Feedback

From a computational perspective, embedding lateral competition within the higher-order feedback loop provides a highly efficient, dynamic mechanism for sensory filtering and predictive coding. Simple additive or multiplicative top-down modulations often lead to runaway excitation or generalized gain changes that lack spatial and feature specificity. In contrast, our simulations indicate that inter-areal competition acts as a highly selective spatial filter. When a visual stimulus extends beyond the classical receptive field, the competitive interactions in V2 ensure that the resulting feedback to V1 is actively sculpted, suppressing redundant neural activity encoding predictable spatial continuities.

This architecture serves two critical computational imperatives. First, it prevents the saturation of primary sensory areas during high-contrast, large-scale visual stimulation, maintaining the network within a stable asynchronous-irregular regime. Second, it profoundly optimizes information transmission; by silencing populations that encode redundant contextual information, the cortex reserves its metabolic and energetic bandwidth to transmit salient features, such as boundaries, discontinuities, or unexpected stimuli. Therefore, the top-down competitive mechanism is not merely an inhibitory reflex, but a sophisticated operation for sparse coding and sensory optimization.

“

5.3 Implications for Machine Learning

Beyond its neuroscientific relevance, the hierarchical competitive mechanism we identify offers a concrete architectural principle that can inform the design of artificial neural networks. The core insight — that lateral inhibition in higher processing stages enables robust, context-dependent normalization — resonates with several recent trends in deep learning, yet remains underexplored as an explicit design choice.

First, our winner-take-all gating of feedback bears functional similarities to *attention mechanisms* (**?**), where queries selectively weight the relevance of different spatial locations or feature channels. However, whereas standard self-attention computes pairwise similarities across all positions — incurring quadratic complexity — our lateral inhibition operates locally between modules, offering a potentially more scalable alternative for spatial normalization. This principle aligns with recent work on *squeeze-and-excitation* networks (**?**), which recalibrate channel-wise features via a lightweight gating mechanism, and could be extended to incorporate lateral competition between feature modules.

Second, the normalization dynamics we observe — where V1 responses are scaled relative to the activity of competing V2 modules — implement a form of *contrastive* computation. Rather than encoding absolute stimulus intensity, the network learns to represent relative differences, a property that has proven essential for self-supervised contrastive learning objectives (**??**). Our biologically constrained architecture suggests that hierarchical lateral inhibition could serve as an energy-efficient alternative to explicit contrastive losses, potentially enabling on-chip learning in neuromorphic hardware (**?**).

Third, the dynamic range compression achieved by competitive feedback offers a potential route to *adversarial robustness*. By normalizing responses relative to spatial context rather than absolute input magnitudes, the network becomes inherently less sensitive to perturbations that uniformly scale input intensities. Preliminary evidence from biology supports this view: surround suppression persists under a wide range of luminance and contrast conditions (**?**), and we hypothesize that artificial systems incorporating similar hierarchical competition may exhibit analogous invariance properties.

Finally, our work demonstrates that these computations can be realized in fully spiking architectures, which are compatible with emerging neuromorphic platforms such as Intel’s Loihi (**?**) and SpiNNaker (**?**). Translating the hierarchical competitive motif to trainable spiking networks — for instance, via surrogate gradient methods (**?**) — represents a promising direction for developing low-power artificial perception systems that inherit the robustness and efficiency of biological sensory processing.”

5.4 Implications for Computational Psychiatry

Beyond basic sensory processing, the mechanistic framework proposed here offers profound insights into the pathophysiology of neuropsychiatric disorders. The delicate balance between excitation and inhibition (E/I balance) required to sustain hierarchical competition in our model is inextricably linked to the generation of high-frequency oscillatory activity. As our results demonstrate, robust contextual suppression is accompanied by prominent gamma-band oscillations ($\sim$ 75 Hz).

When we systematically perturbed this local E/I balance to simulate pathological endophenotypes—specifically by attenuating superficial inhibitory synaptic efficacy—the network exhibited a severe degradation in surround suppression alongside a distinct spectral shift and reduction in gamma power. This neurodynamic signature closely mirrors the sensory gating deficits and aberrant oscillatory profiles frequently observed in patients with schizophrenia. Consequently, our spiking network model transitions from a purely theoretical construct to a powerful *in silico* translational platform for computational psychiatry. It provides a cohesive, circuit-level explanation for how localized synaptic micro-deficits can propagate through macroscopic cortical hierarchies, ultimately manifesting as well-documented clinical endophenotypes.

5.5 Limitations and Future Directions

While our model successfully captures the core non-linear dynamics of eCRF modulation, several avenues remain for future theoretical and empirical investigation. The current implementation relies on computationally efficient, simplified point-neuron models (Leaky Integrate-and-Fire) and utilizes generalized inhibitory population pools. However, cortical inhibition is highly diverse. Future iterations will significantly benefit from disentangling specific interneuron subtypes. For instance, Parvalbumin- expressing (PV) interneurons are primarily implicated in fast somatic feedforward inhibition and the pacing of gamma oscillations, whereas Somatostatin-expressing (SST) interneurons preferentially target distal dendrites, receive dense top-down feedback, and are theorized to directly mediate surround suppression.

Furthermore, the present model operates with static synaptic weights. Incorporating spike-timing-dependent plasticity (STDP) or other biologically plausible learning rules could elucidate how these hierarchical competitive networks develop their precise topographic connectivity during visual maturation. Finally, exploring how these network dynamics adapt under varying neuromodulatory states (e.g., cholinergic or glutamatergic modulation) will further clarify the biological constraints of top-down sensory processing in both health and disease.

“In summary, this work demonstrates that a specific circuit motif — lateral inhibitory competition in higher cortical areas — is both necessary and sufficient to explain the hallmark signature of visual context integration: robust surround suppression. By grounding this principle in a large-scale spiking microcircuit, we provide a mechanistic bridge between decades of physiological data and the computational principles of hierarchical normalization. The resulting architecture not only advances our understanding of cortical computation but also offers a tangible blueprint for the next generation of robust, energy-efficient artificial perception systems.”

6 Submission of papers to NeurIPS 2026

Please read the instructions below carefully and follow them faithfully.

6.1 Style

Papers to be submitted to NeurIPS 2026 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. **Papers that exceed the page limit will not be reviewed (or in any other way considered) for presentation at the conference.** Additional pages *containing acknowledgments, references, checklist, and optional technical appendices* do not count as content pages.

The margins in 2026 are the same as those in previous years.

Authors are required to use the NeurIPS LaTeX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for desk rejection.

6.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

`https://neurips.cc`.

The only supported style file for NeurIPS 2026 is `neurips_2026.sty`, rewritten for LaTeX 2_{ε}. **Previous style files for LaTeX 2.09, Microsoft Word, and RTF are no longer supported.**

The LaTeX style file contains three optional arguments:

- `final`, which creates a camera-ready copy,
- `preprint`, which creates a preprint for submission to, e.g., arXiv,
- `nonatbib`, which will not load the `natbib` package for you in case of package clash.

Preprint option If you wish to post a preprint of your work online, e.g., on arXiv, using the NeurIPS style, please use the `preprint` option. This will create a nonanonymized version of your work with the text "Preprint. Work in progress." in the footer. This version may be distributed as you see fit, as long as you do not say which conference it was submitted to. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

At submission time, please omit the `final` and `preprint` options. This will anonymize your submission and add line numbers to aid review. Please do *not* refer to these line numbers in your paper as they will be removed during generation of camera-ready copies.

The file `neurips_2026.tex` may be used as a "shell" for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 7, 8, and 9 below.

7 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 9 regarding figures, tables, acknowledgments, and references.

8 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

8.1 Headings: second level

Second-level headings should be in 10-point type.

8.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

9 Citations, figures, tables, references

These instructions apply to everyone.

9.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated...

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2026` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2026}
```

As submission is double blind, refer to your own published work in the third person. That is, use "In the previous work of Jones et al. [4]," not "In our previous work [4]." If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form "A. Anonymous" and include a copy of the anonymized paper in the supplementary material.

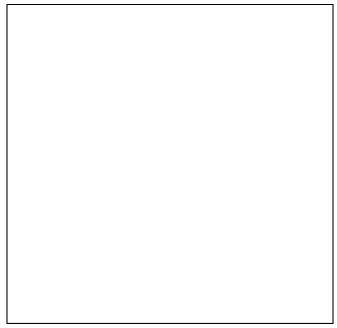

Figure 2: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

Table 2: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	$\approx$100
Axon	Output terminal	$\approx$10
Soma	Cell body	up to 10^6

9.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[2] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[3]

9.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for the first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

9.4 Tables

All tables must be centered, neat, clean, and legible. The table number and title always appear before the table. See Table 2.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for the first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules.* We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 2.

[2] Sample of the first footnote.
[3] As in this example.

9.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using $$ anyway; see `https://tex.stackexchange.com/questions/503/why-is-preferable-to` and `https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath` for more information.)

9.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes. Please note that pages should be numbered.

10 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

10.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Technical appendices and supplementary material

Technical appendices with additional results, figures, graphs, and proofs may be submitted with the paper submission before the full submission deadline (see above). You can upload a ZIP file for videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the technical appendices.

Note: Think of the appendix as "optional reading" for reviewers. The paper must be able to stand alone without the appendix; for example, adding critical experiments that support the main claims to an appendix is inappropriate.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: **[TODO]**

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- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: **[TODO]**

   Justification: **[TODO]**

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10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
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    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: **[TODO]**

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- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
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    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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    Justification: **[TODO]**

    Guidelines:

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    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

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    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

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